

CUTTING DATA

Parameters based on ideal conditions. Please adjust parameters accordingly to real applications.

● Recommended
○ Suitable

Parameters based on ideal conditions.
Please adjust parameters accordingly to real applications.

	Material	Ø	Vc m/min	1	1.5	2	2.5	3	4	5	6	8	10	12	16	20
				fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)	fz (mm/tooth)
P	● 1.1 Free Cutting Steel	< 120 HB	158	0.012-0.021	0.015-0.024	0.017-0.028	0.018-0.030	0.019-0.031	0.023-0.038	0.027-0.045	0.033-0.055	0.045-0.075	0.054-0.090	0.062-0.103	0.076-0.126	0.086-0.143
	● 1.2 Structural Steel	< 200 HB	153	0.012-0.021	0.015-0.024	0.017-0.028	0.018-0.030	0.019-0.031	0.023-0.038	0.027-0.045	0.033-0.055	0.045-0.075	0.054-0.090	0.062-0.103	0.076-0.126	0.086-0.143
	● 1.3 Plain Carbon Steel	< 250 HB	135	0.008-0.014	0.010-0.017	0.012-0.020	0.014-0.023	0.016-0.026	0.019-0.032	0.023-0.038	0.027-0.045	0.034-0.056	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
	● 1.4 Alloy Steel	< 250 HB	125	0.008-0.014	0.008-0.014	0.008-0.014	0.009-0.016	0.011-0.018	0.017-0.029	0.020-0.034	0.025-0.041	0.034-0.056	0.041-0.068	0.047-0.078	0.057-0.095	0.065-0.108
	● 1.5 Low Alloy Steel	250 - 350 HB	125	0.008-0.014	0.008-0.014	0.008-0.014	0.009-0.016	0.011-0.018	0.017-0.029	0.020-0.034	0.025-0.041	0.034-0.056	0.041-0.068	0.047-0.078	0.057-0.095	0.065-0.108
	● 1.6 Low Alloy Steel	> 350 HB	120	0.008-0.014	0.008-0.014	0.008-0.014	0.009-0.016	0.011-0.018	0.017-0.029	0.020-0.034	0.025-0.041	0.034-0.056	0.041-0.068	0.047-0.078	0.057-0.095	0.065-0.108
M	● 2.1 Free Machining Stainless Steel	< 250 HB	97	0.009-0.016	0.010-0.017	0.011-0.018	0.013-0.021	0.015-0.025	0.019-0.032	0.023-0.038	0.027-0.045	0.037-0.061	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
	● 2.2 Austenitic Stainless Steel	< 320 HB	97	0.009-0.016	0.010-0.017	0.011-0.018	0.013-0.021	0.015-0.025	0.019-0.032	0.023-0.038	0.027-0.045	0.037-0.061	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
	● 2.3 Ferritic and Martensitic Stainless Steel	< 300 HB	75	0.006-0.009	0.008-0.013	0.010-0.016	0.010-0.017	0.011-0.018	0.013-0.022	0.016-0.026	0.019-0.031	0.026-0.043	0.031-0.051	0.035-0.059	0.044-0.073	0.049-0.081
	● 2.4 Precipitation Hardened Stainless Steel	320 - 410 HB	72	0.006-0.009	0.008-0.013	0.010-0.016	0.010-0.017	0.011-0.018	0.013-0.022	0.016-0.026	0.019-0.031	0.026-0.043	0.031-0.051	0.035-0.059	0.044-0.073	0.049-0.081
	● 3.1 Lamellar Graphite Cast Iron	< 150 HB	135	0.012-0.021	0.015-0.024	0.017-0.028	0.018-0.030	0.019-0.031	0.023-0.038	0.027-0.045	0.033-0.055	0.045-0.075	0.054-0.090	0.062-0.103	0.076-0.126	0.086-0.143
	● 3.2 Lamellar Graphite Cast Iron	150 - 300 HB	130	0.012-0.021	0.015-0.024	0.017-0.028	0.018-0.030	0.019-0.031	0.023-0.038	0.027-0.045	0.033-0.055	0.045-0.075	0.054-0.090	0.062-0.103	0.076-0.126	0.086-0.143
K	● 3.3 Nodular Graphite, Malleable Cast Iron	< 200 HB	135	0.008-0.014	0.010-0.017	0.012-0.020	0.014-0.023	0.016-0.026	0.019-0.032	0.023-0.038	0.027-0.045	0.034-0.056	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
	● 3.4 Nodular Graphite, Malleable Cast Iron	< 200 HB	128	0.008-0.014	0.008-0.014	0.008-0.014	0.009-0.016	0.011-0.018	0.015-0.026	0.018-0.030	0.023-0.038	0.031-0.051	0.037-0.061	0.042-0.070	0.052-0.086	0.058-0.096
	● 4.1 Titanium unalloyed	< 200 HB	65	0.008-0.014	0.009-0.015	0.009-0.016	0.011-0.018	0.013-0.022	0.015-0.026	0.018-0.030	0.023-0.038	0.031-0.051	0.037-0.061	0.042-0.070	0.052-0.086	0.058-0.096
	● 4.2 Titanium alloyed	< 270 HB	57	0.008-0.014	0.008-0.014	0.008-0.014	0.011-0.018	0.013-0.022	0.015-0.026	0.018-0.030	0.023-0.038	0.031-0.051	0.037-0.061	0.042-0.070	0.052-0.086	0.058-0.096
	● 4.3 Titanium alloyed	270 - 350 HB	50	0.008-0.014	0.008-0.014	0.008-0.014	0.011-0.018	0.013-0.022	0.015-0.026	0.018-0.030	0.023-0.038	0.031-0.051	0.037-0.061	0.042-0.070	0.052-0.086	0.058-0.096
	● 5.1 Nickel unalloyed	< 150 HB	50	0.011-0.018	0.012-0.020	0.013-0.022	0.014-0.024	0.015-0.026	0.019-0.032	0.023-0.038	0.027-0.045	0.037-0.061	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
Ni	● 5.2 Nickel alloyed	< 270 HB	45	0.011-0.018	0.012-0.020	0.013-0.022	0.014-0.024	0.015-0.026	0.019-0.032	0.023-0.038	0.027-0.045	0.037-0.061	0.044-0.074	0.050-0.084	0.062-0.104	0.071-0.118
	● 5.3 Nickel alloyed	270 - 350 HB	27	0.007-0.011	0.008-0.013	0.008-0.014	0.009-0.016	0.011-0.018	0.015-0.026	0.018-0.030	0.023-0.038	0.031-0.051	0.037-0.061	0.042-0.070	0.052-0.086	0.058-0.096
	○ 6.1 Copper	< 100 HB	240	0.011-0.019	0.016-0.026	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 6.2 Beta Brass, Bronze	< 200 HB	230	0.011-0.019	0.016-0.026	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 6.3 Alpha Brass	< 200 HB	230	0.011-0.019	0.016-0.026	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 6.4 High Strength Bronze	< 470 HB	180	0.013-0.022	0.017-0.028	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
Cu	○ 7.1 Aluminium Magnesium unalloyed	< 100 HB	438	0.013-0.022	0.017-0.028	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 7.2 Aluminium Alloy < 5% Si	< 150 HB	178	0.013-0.022	0.017-0.028	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 7.3 Aluminium Alloy 5 to 10% Si	< 120 HB	110	0.013-0.022	0.017-0.028	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 7.4 Aluminium Alloy > 10% Si	< 120 HB	100	0.013-0.022	0.017-0.028	0.020-0.033	0.023-0.038	0.026-0.043	0.032-0.053	0.038-0.063	0.053-0.088	0.063-0.100	0.060-0.100	0.060-0.100	0.068-0.113	0.075-0.125
	○ 8.1 Duplexsteels (short chipping)	< 240 HB	240	0.006-0.009	0.007-0.012	0.008-0.014	0.009-0.015	0.009-0.016	0.011-0.019	0.014-0.023	0.017-0.028	0.023-0.038	0.027-0.045	0.033-0.055	0.044-0.073	0.055-0.091
	○ 8.2 Thermoplastics (long chipping)	< 150 HB	235	0.006-0.009	0.007-0.012	0.008-0.014	0.009-0.015	0.009-0.016	0.011-0.019	0.014-0.023	0.017-0.028	0.023-0.038	0.027-0.045	0.033-0.055	0.044-0.073	0.055-0.091
Syn	○ 8.3 Fibre Reinforced Synthetic Materials		95	0.006-0.009	0.007-0.012	0.008-0.014	0.009-0.015	0.009-0.016	0.011-0.019	0.014-0.023	0.017-0.028	0.023-0.038	0.027-0.045	0.033-0.055	0.044-0.073	0.055-0.091

● Recommended
○ Suitable

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Please adjust parameters accordingly to real applications.

ap ap = 1.5 x d ae = 0.2 x d Side Milling = fz Vc ap ae ap = 0.5 x d ae = 1.0 x d Slot Milling = fz x 0.7		Material																
		Ø	1	1.5	2	2.5	3	4	5	6	8	10	12	16	20	Recommended		
P	● 1.1 Free Cutting Steel	< 120 HB	158	0.009-0.016	0.012-0.020	0.015-0.025	0.018-0.030	0.021-0.034	0.033-0.055	0.045-0.075	0.056-0.094	0.064-0.106	0.075-0.125	0.075-0.125	0.090-0.150	0.090-0.150	○ Suitable	
	● 1.2 Structural Steel	< 200 HB	153	0.009-0.016	0.012-0.020	0.015-0.025	0.018-0.030	0.021-0.034	0.033-0.055	0.045-0.075	0.056-0.094	0.064-0.106	0.075-0.125	0.075-0.125	0.090-0.150	0.090-0.150	○ Suitable	
	● 1.5 Low Alloy Steel	250 - 350 HB	125	0.009-0.016	0.009-0.016	0.009-0.016	0.013-0.022	0.017-0.028	0.023-0.038	0.030-0.050	0.045-0.075	0.060-0.100	0.068-0.113	0.075-0.125	0.083-0.138	0.083-0.138	○ Suitable	
	● 1.6 Low Alloy Steel	> 350 HB	120	0.009-0.016	0.009-0.016	0.009-0.016	0.013-0.022	0.017-0.028	0.023-0.038	0.030-0.050	0.045-0.075	0.060-0.100	0.068-0.113	0.075-0.125	0.083-0.138	0.083-0.138	○ Suitable	
	● 2.2 Austenitic Stainless Steel	< 320 HB	97	0.006-0.009	0.008-0.013	0.009-0.016	0.010-0.017	0.011-0.019	0.017-0.028	0.023-0.038	0.030-0.050	0.034-0.056	0.045-0.075	0.049-0.081	0.060-0.100	0.064-0.106	○ Suitable	
M	● 2.4 Precipitation Hardened Stainless Steel	320 - 410 HB	72	0.006-0.009	0.006-0.009	0.006-0.009	0.008-0.013	0.009-0.016	0.014-0.023	0.018-0.030	0.023-0.038	0.026-0.044	0.032-0.053	0.038-0.063	0.056-0.094	0.064-0.106	○ Suitable	
	● 3.1 Lamellar Graphite Cast Iron	< 150 HB	135	0.009-0.016	0.012-0.020	0.015-0.025	0.018-0.030	0.021-0.034	0.033-0.055	0.045-0.075	0.056-0.094	0.064-0.106	0.075-0.125	0.075-0.125	0.090-0.150	0.090-0.150	○ Suitable	
	● 3.2 Lamellar Graphite Cast Iron	150 - 300 HB	130	0.009-0.016	0.012-0.020	0.015-0.025	0.018-0.030	0.021-0.034	0.033-0.055	0.045-0.075	0.056-0.094	0.064-0.106	0.075-0.125	0.075-0.125	0.090-0.150	0.090-0.150	○ Suitable	
K	● 3.3 Nodular Graphite, Malleable Cast Iron	< 200 HB	135	0.009-0.016	0.009-0.016	0.009-0.016	0.013-0.022	0.017-0.028	0.023-0.038	0.030-0.050	0.045-0.075	0.060-0.100	0.068-0.113	0.075-0.125	0.083-0.138	0.083-0.138	○ Suitable	
	● 3.4 Nodular Graphite, Malleable Cast Iron	200 - 320 HB	128	0.009-0.016	0.009-0.016	0.009-0.016	0.013-0.022	0.017-0.028	0.023-0.038	0.030-0.050	0.045-0.075	0.060-0.100	0.068-0.113	0.075-0.125	0.083-0.138	0.083-0.138	○ Suitable	
Ti	● 4.1 Titanium unalloyed	< 200 HB	65	0.008-0.014	0.009-0.015	0.010-0.017	0.012-0.019	0.013-0.022	0.015-0.025	0.019-0.031	0.023-0.038	0.026-0.044	0.034-0.056	0.038-0.063	0.060-0.100	0.075-0.125	○ Suitable	
Ni	● 5.1 Nickel unalloyed	< 150 HB	50	0.006-0.009	0.007-0.012	0.008-0.014	0.009-0.015	0.009-0.016	0.011-0.019	0.014-0.023	0.017-0.028	0.023-0.038	0.027-0.045	0.033-0.055	0.044-0.073	0.055-0.091	○ Suitable	
H	● 9.1 Hardened Steel	< 32 HRC	120	0.006-0.009	0.008-0.013	0.009-0.016	0.012-0.020	0.015-0.025	0.023-0.038	0.030-0.050	0.045-0.075	0.060-0.100	0.068-0.113	0.075-0.125	0.083-0.138	0.083-0.138	○ Suitable	
	● 9.2 Hardened Steel	33 - 41 HRC	95	0.006-0.009	0.006-0.009	0.006-0.009	0.008-0.013	0.009-0.016	0.019-0.031	0.026-0.044	0.038-0.063	0.049-0.081	0.053-0.088	0.059-0.100	0.060-0.100	0.060-0.100	○ Suitable	
	● 9.3 Hardened Steel	42 - 50 HRC	90	0.006-0.009	0.006-0.009	0.006-0.009	0.008-0.013	0.009-0.016	0.019-0.031	0.026-0.044	0.038-0.063	0.049-0.081	0.053-0.088	0.059-0.100	0.060-0.100	0.060-0.100	○ Suitable	